IRÈNE HU

Environmental Economist at the OECD
Ph.D. from the Paris School of Economics
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RESEARCH INTERESTS

Primary fields	Environmental Economics, Development Economics
Secondary fields	Migration Economics, Health Economics, Economics of Education

WORK EXPERIENCE

OECD Since May 2024
Environmental economist (Environment Directorate) *Paris, France*

- I provide economic, policy analysis and advice in research questions on environmental and economic policy integration to address the triple planetary crisis of climate change, biodiversity loss and pollution (recently published Environmental Outlook report).
- I work on quantitative and econometric methods to estimate the valuation of adverse health impacts of chemicals through WTP surveys (SWACHE project).

Jan 2023 – May 2024
Analyst, internal consultant (Education Directorate) *Paris, France*

- Data analysis and drafting of the PISA 2022 Initial Report Volume II on the resilience of education systems, schools and students during and after the Covid-19 pandemic, the country note for France and a PISA in Focus note entitled "Managing screen time: How to protect and equip students against distraction".
- Presented the PISA 2022 results for France to 50 journalists at the official embargoed media briefing

World Bank Oct 2021 – Sep 2022
Consultant (STC), Poverty and Equity Global Practice, Europe and Central Asia *Paris, France*

- Conducted statistical analyses of the World Bank Pulse surveys to measure the socio-economic and labour impact of Covid-19 in Bulgaria, Croatia, Poland and Romania
- Contributed to the EU Regular Economic Report (RER) 8 and the Romania Systematic Country Diagnostic (SCD) Update.

World Health Organization Jun – Sep 2017
Research assistant *Ulaanbaatar, Mongolia*

- Conducted an evaluation of the health and economic impacts of an excise tax on sugar-sweetened beverages in Mongolia

World Bank Development Research Group (DECRG) Sep 2015 – Aug 2016
Research assistant to Quy-Toan Do and Hanan Jacoby *Washington D.C., USA*

- Data analysis and literature review for a randomized pricing experiment aimed at assessing the willingness-to-pay for piped water in Vietnam (Optimal Pricing of a New Utility Service: The Case of Piped Water in Vietnam, accepted at the Review of Economic Studies)

Abdul Latif Jameel Poverty Action Lab (J-PAL Europe) Nov–Dec 2014
Field investigator for Vincent Pons *Paris, France*

- Administered door-to-door questionnaires in the suburb of Courcouronnes for an impact evaluation on the access to health care among underprivileged households.

United Nations Secretariat Jul – Aug 2014
Research intern to David Leblanc, Division for Sustainable Development (DESA) *New York City, USA*

- Carried out a comparative analysis of integrated climate, land, energy and water policies in twenty countries.

EDUCATION

Paris School of Economics *Sept. 2018 - March 2023*
Ph.D. in Economics
Title: *Three Essays on the Industrial Exploitation of Natural Resources in Africa*
Supervisors: Fabrice Etilé (PSE, INRAE) and François Libois (PSE, INRAE)
Jury: Jérémy Laurent-Lucchetti (University of Geneva), Katrin Millock (PSE, CNRS), Céline Nauges (TSE, INRAE), Raphaël Soubeyran (INRAE, CEE-Montpellier)

Paris School of Economics *2016-2018*
Masters in Analysis and Policy in Economics

ENSAE ParisTech (French National School of Statistics and Economics) *2013-2017*
Engineering Masters' Degree in Data Science, Quantitative Economics and Sociology

Lycée Henri IV, Paris *2011-2013*
Classes Préparatoires aux Grandes Écoles, Economics and Business (ECS)

PUBLICATIONS

Valuing a reduction in the risk of miscarriage: A large-scale multi-country stated preference approach *OECD Environment Working Papers, No. 271, OECD Publishing, Paris, 2026*
with Chris Dockins, Damien Dussaux, Chau-Man Fung, Stavros Georgiou, Charles Griffiths and Nathalie Simon.

Valuing a reduction in the risk of an underactive thyroid: A large-scale multi-country stated preference approach *OECD Environment Working Papers, No. 272, OECD Publishing, Paris, 2026*
with Heini Ahtiainen, Richard Dubourg, Damien Dussaux, Väinö Nurmi, Elvia Rufo Jimenez and Christoph Rheinberger.

Environmental Outlook on the Triple Planetary Crisis: Stakes, Evolution and Policy Linkages *OECD Publishing, Paris, 2025*
led by Shardul Agrawala.

Managing screen time: How to protect and equip students against distraction, *PISA in Focus, No. 124, OECD Publishing, Paris, 2024*
with Andreea Minea-Pic.

PISA 2022 results Volume II - Learning During – and From – Disruption, *PISA, OECD Publishing, Paris, 2023*
led by Miyako Ikeda, with Hannah Ulferts and Alfonso Echazarra.

Programme International pour le suivi des acquis des élèves (PISA) : Principaux résultats pour la France du PISA 2022 *PISA, Éditions OCDE, Paris, 2023*
with Éric Charbonnier.

Romania - Systematic Country Diagnostic Update, *World Bank, 2023,* Washington, D.C. : World Bank Group.
led by Nga Thi Viet Nguyen and Emilija Timmis.

EU Regular Economic Report 8: Living Up to Potential in the Wake of Adverse Shocks - Part 1, *World Bank report on the European Union, 2022*
with Mona Prasad, Emilija Timmis, Collette Mari Wheeler, Andrei Silviu Dospinescu, Kristina Vaughan, Samuel Freije-Rodriguez, Nga Thi Viet Nguyen, Reena Badiani-Magnusson, Leonardo Iacovone, Lukasz Marek Marc, Natasha Kapil and Matija Laco.

Prices, Patents and Access to Drugs: Views on Equity and Efficiency in the Global Pharmaceutical Industry, *Revue française des affaires sociales 2018/3*
with Quentin Cavalan, Maud Hazan and Roxane Zighed.

Benefits of Fiscal Policies on Sugar-Sweetened Beverages, *Central Asian Journal of Medical Sciences, 2017*
with Undram Mandakh, Erdenechimeg Erdenebayar, Khorolmaa Bold, Tseden Purevdorj and Chimedsuren Ochir.

WORKING PAPERS

MiningLeaks: Water Pollution and Child Mortality in Africa, *with Mélanie Gittard*
[Under review]

In the midst of Africa's mining boom, communities downstream from industrial mines face increased exposure to toxic waste. Yet, the effects of induced water pollution on the local population's health have not been quantified at the continental scale of Africa, due to data limitation and non-random exposure. This paper investigates this question using a quasi-experimental design and a novel dataset detailing the location and opening dates of all known industrial mines, obtained through intensive manual data collection. We combine geo-coded information on 2,016 industrial mines with health outcomes from the Demographic Health Survey (DHS) from 1986 to 2018 in 26 African countries. Through a staggered difference-in-difference strategy, we compare villages downstream and upstream of mines before and after opening and find a 25% increase in 24-month mortality rates downstream. The effect is mainly observed among children who were no longer breastfed, confirming that water pollution drives the results. Our analysis rules out other mechanisms like fertility changes, access to facilities, in-migration, conflicts and income effects. The impact intensifies during mine operation and high international mineral prices, especially in densely mined regions, and fades out with distance.

Man Overboard! Industrial Fishing as a Driver of Migration Out of Africa, *with François Libois*

The adoption of a capital intensive technology to extract natural resource, competing directly with a more traditional and labour intensive use, can trigger migration in developing countries. This article focuses on the effect of fish stock depletion on migration in Africa and uses a novel dataset on industrial fishing intensity (Kroodsma et al., 2018). Based on a panel of the 37 African countries with access to the sea over the period 2012-2018, we show that within country variation in industrial fishing intensity increases migration to OECD countries. We find strong evidence that the competition created by industrial fishing vessels overfishing African seas and depleting fish stocks, increase the number of migrants to OECD countries. We then show that findings at the macro level are consistent in terms of mechanisms with micro level estimates using household level demographic data.

Raw Materials Diplomacy, Official Development Finance and the Industrial Exploitation of Natural Resources in Africa
(draft available upon request)

This paper investigates how official development finance from traditional and emerging donors influences the industrial exploitation of natural resources in Africa. Willingly or not, loans contribute to the local influence a donor country can have in a recipient country, and its companies can benefit from easier access to extract the recipient's natural resources. Using development finance flows panel data from the OECD countries,

China, India and the Gulf countries on the one side, and novel data on industrial extraction of natural resources on the other side. Through an extended descriptive analysis, I find that official development flows from DAC and non-DAC countries are positively correlated with their capacity to conclude large-scale land acquisitions in Africa, over 2000-2014.

GRANTS AND AWARDS

Full doctoral scholarship, Université Paris 1 - Panthéon Sorbonne	*2018-2022*
Research grant, CEPREMAP (5k)	*2022*
Research grant, EUR-PGSE (2.5k)	*2022*
Best PhD project, IOEA	*2019*
Research grant, EUR-PGSE (4.5k)	*2019*
International mobility grant, Université Paris-Saclay (1.9k)	*2017*

SEMINARS AND CONFERENCES

2023 13th Annual Conference on Immigration in OECD Countries
European Economics Association (EEA-ESEM) Annual Conference

2022 European Health Economics Association (EuHEA) Seminar Series
European Association of Environmental and Resource Economists (EAERE) Annual Conference
Journées doctorales du développement (JDD, Centre d'Économie de la Sorbonne)
Casual Friday Development Seminar (CFDS, PSE)

2021 Center for the Studies of African Economies (CSAE) Annual Conference
69th Congress of the French Economic Association (AFSE)
19th Nordic Conference in Development Economics (NCDE)
International Development Economics Conference (ICDE)
14th Int. Conference on Migration and Development (LISER, U. of Luxembourg, AFD, World Bank)
Development Finance and Public Policies (DeFiPP) Workshop (University of Namur)
Casual Friday Development Seminar (CFDS, PSE)

2020 EAERE-ETH European Winter School
Center for the Studies of African Economies (CSAE) Annual Conference
7th French Association of Environmental and Resource Economics (FAERE) Annual Conference
Applied Economics Lunch Seminar (PSE)
Casual Friday Development Seminar (CFDS, PSE)
2nd International Conference on Environmental Economics, A Focus on Natural Resources (EENR)

2019 Institutional and Organizational Economics Academy (IOEA), 18th Edition Spring School
Conférence Internationale de Statistique et d'Economie Appliquée (CISEA, ENSEA Abidjan)
Casual Friday Development Seminar (CFDS, PSE)

ADDITIONAL ACTIVITIES

Teaching Teaching Assistant in Development Economics for Golvine de Rochambeau, Fall 2020
Master 1 (Paris School of International Affairs) and Master 2 (School of Public Affairs)

Refereeing Environment and Development Economics Journal, World Development

SKILLS

Programming and software	Stata, R, QGIS, Python
Languages	French (native) English (fluent) Mandarin Chinese (fluent, HSK level 5) German (intermediate)

REFERENCES

Quy-Toan DO
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Co-Director for the 2023 World Development Report
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Policy Evaluation, Social and
Distributional Outcomes (ENV/EEI)
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